

Databricks Dashboard

(<https://dbc-bf4adfc2-54bd.cloud.databricks.com/dashboardsv3/01f18b51ec271049b74790d98fed7adc/published?o=7474660577300899>)

1. Project Overview

This project analyzes historical sales data to evaluate product performance, pricing strategies, profitability, and customer demand. Using SQL in Databricks, I developed key business metrics and analytical models to identify sales trends, promotional periods, and price elasticity of demand. Thereafter, I used Databricks Dashboards to create visualisations to create a dashboard. Please see screenshots below.

This project demonstrates my ability to transform raw transactional data into meaningful business insights that support pricing, promotional, and operational decision-making.

2. Business Objectives

The analysis was conducted to answer the following business questions:

- What is the daily sales price per unit?
- What is the average unit sales price?
- What is the daily gross profit percentage?
- What is the daily gross profit per unit?
- Which promotional periods can be identified from the data?
- What is the Price Elasticity of Demand during these promotional periods?
- Does the product perform better when sold at a promotional price?

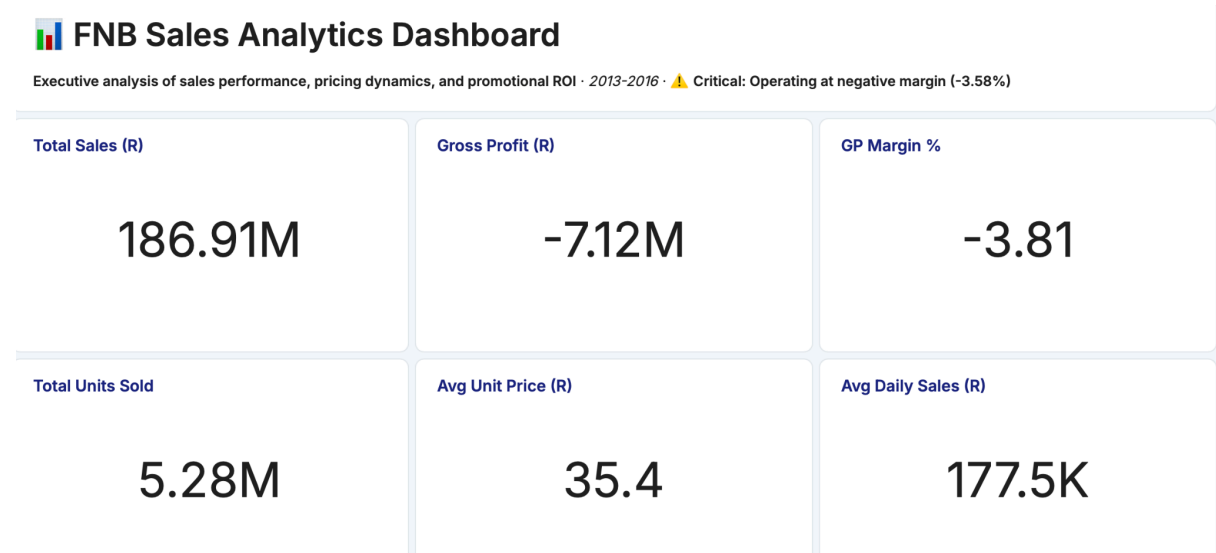
3. Dataset Overview

Dataset Information

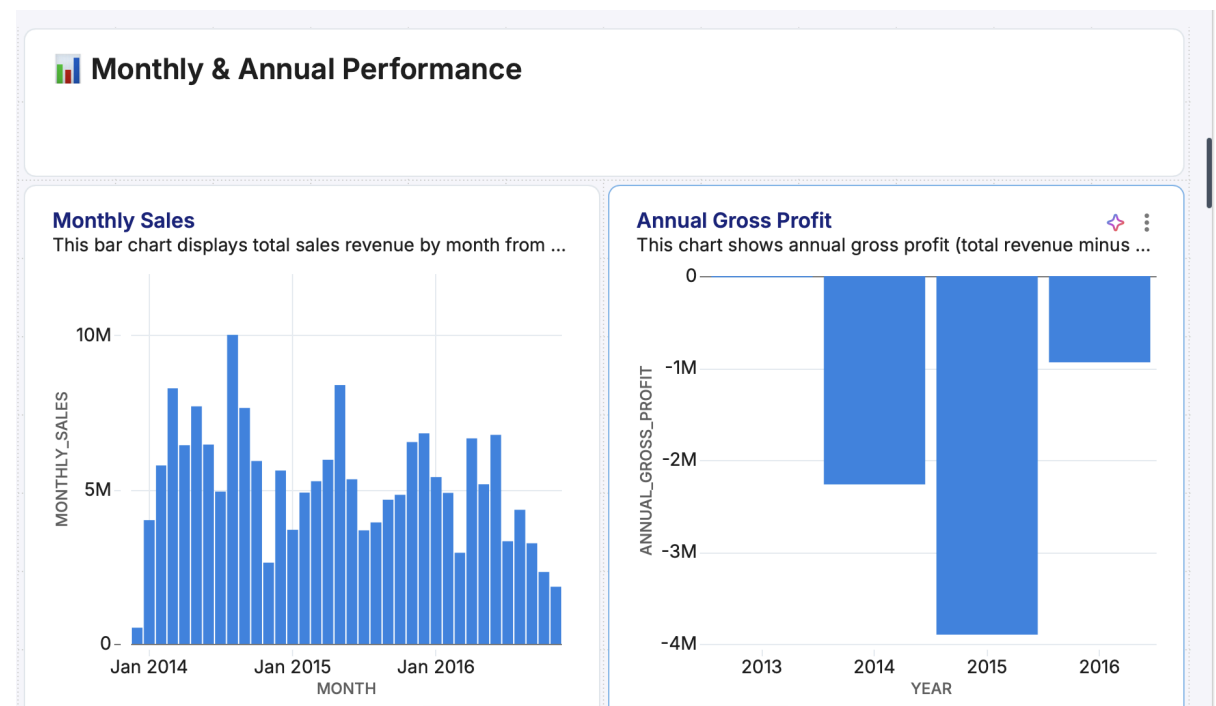
- Source
- Number of records
- Date range
- Key fields

4. Dashboard Screenshots

4.1 Key Metrics



4.2 Annual Sales Performance

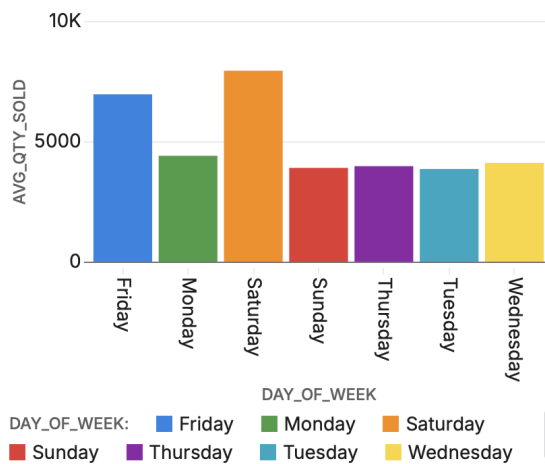


4.3 Weekly Sales Performance

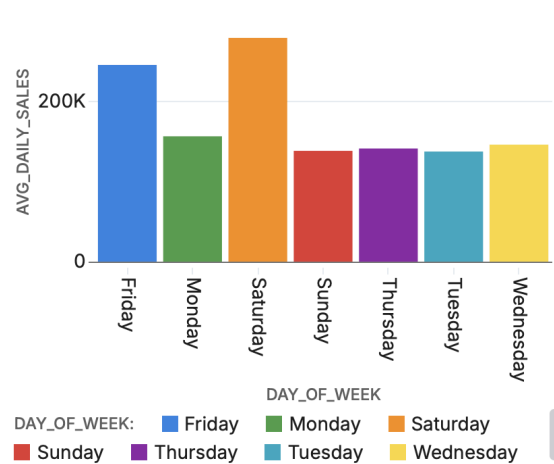
Weekly Sales Patterns

Key Finding: Saturday and Friday are peak sales days, with volume **2x higher** than weekdays. Optimize promotional campaigns and inventory planning around these weekly patterns.

Weekly Sales Pattern - Avg Quantity by Day of W...
This chart reveals clear weekly seasonality: Saturday is th...



Avg Daily Sales by Day of Week
This chart shows average revenue by day of week, mirrori...



4.4 Promotional Period Analysis

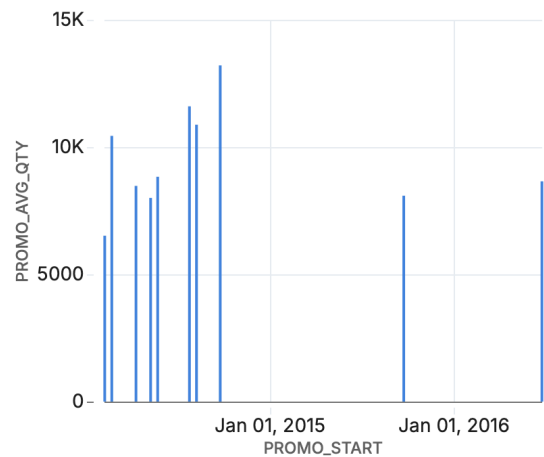
Promotional Campaigns Summary

This table lists all detected promotional campaigns (5+ co...

PROMO_START	PROMO_END	PROMO...
2014-05-23 12:00 AM	2014-06-23 12:00 AM	
2014-08-08 12:00 AM	2014-09-08 12:00 AM	
2014-02-21 12:00 AM	2014-03-17 12:00 AM	
2016-06-23 12:00 AM	2016-07-06 12:00 AM	
2015-09-23 12:00 AM	2015-10-06 12:00 AM	
2014-05-09 12:00 AM	2014-05-21 12:00 AM	
2014-04-10 12:00 AM	2014-04-22 12:00 AM	
2014-09-24 12:00 AM	2014-10-06 12:00 AM	
2014-07-25 12:00 AM	2014-08-05 12:00 AM	
2014-02-07 12:00 AM	2014-02-17 12:00 AM	

Promotional Volume Impact

This chart shows the average daily quantity sold during ea...

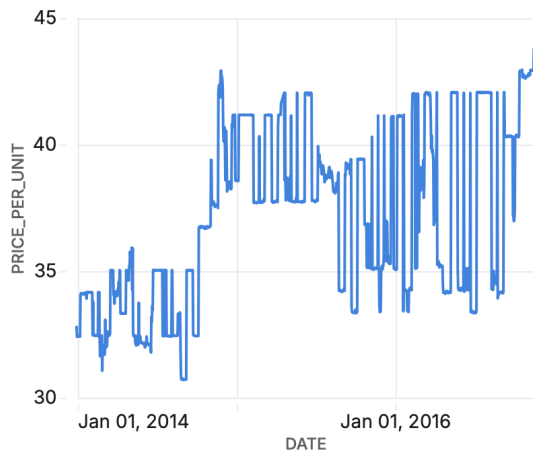


4.5 Profit and Price Trends

Price & Profit Trends

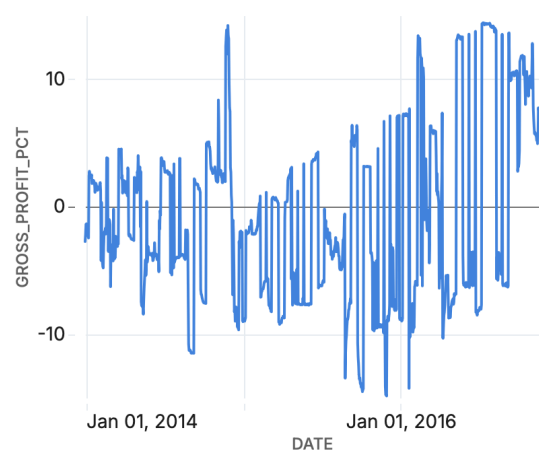
Daily Price Trend

This line chart tracks the daily average price per unit over ...



Daily Gross Profit %

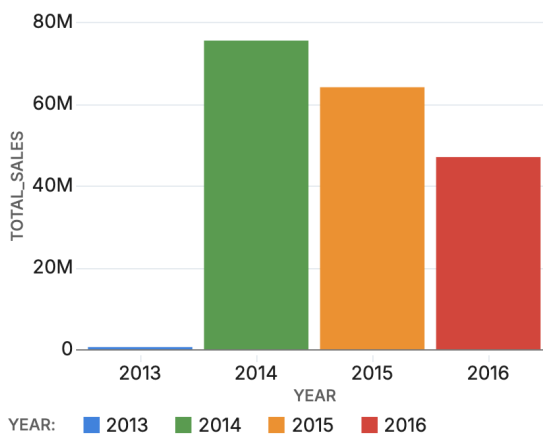
This chart reveals a critical profitability crisis: gross profit ...



4.6 Year-over-Year Performance

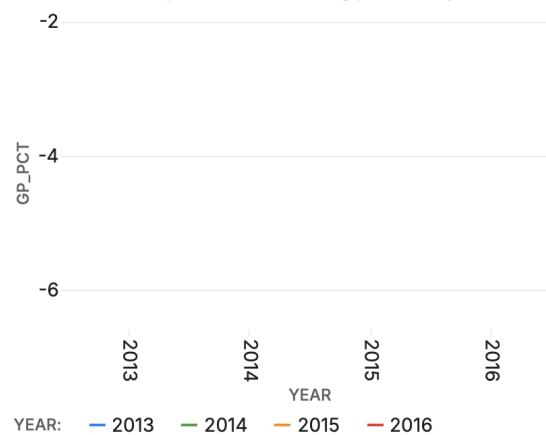
Year-over-Year Sales Comparison

This chart visualizes the dramatic revenue decline over thr...



Year-over-Year GP% Trend

This line chart exposes the worsening profitability crisis: 2...



4.7 Key Findings

CRITICAL FINDINGS & STRATEGIC RECOMMENDATIONS



URGENT: Negative Gross Profit Crisis

Current GP Margin: -3.58% | The product is being sold at a **loss** — costs exceed revenue across the entire period.

Root Cause Analysis:

- **2013-2016 Loss Pattern:** All years show negative margins (-2.4% to -6.1%)
- **Worst Period:** 2015 had the steepest loss at **-6.09% GP margin**
- **Extreme Days:** Late 2015 saw margins as low as **-15%** (costs 15% higher than sales prices)

Key Opportunity: Promotional Strategy is Highly Effective

Price Elasticity Evidence:

- **Period 1:** 13% price cut → **76% volume increase** (PED: -5.89)
- **Period 2:** 14% price cut → **117% volume increase** (PED: -8.38)
- **Period 3:** 13% price cut → **108% volume increase** (PED: -8.22)

Business Impact:  Promotional days average **8,800-10,900 units/day**

 Baseline (non-promo) days average only **5,014 units/day**

 **Promotional uplift: +76% to +117%** volume increase

4.8 Business Recommendations

IMMEDIATE ACTION ITEMS

1. Fix Cost Structure (Priority #1)

-  Investigate why Cost of Sales exceeds revenue
- Review supplier contracts and negotiate better terms
- Conduct product cost audit to identify inefficiencies
- **Target:** Achieve positive GP margin within 90 days

2. Pricing Strategy Review

- Current avg price: R35.40/unit may be too low given costs
- Consider **strategic price increase** of 10-15% on baseline (non-promo) days
- Maintain promotional pricing since it drives massive volume

3. Optimize Promotional Calendar

- **Weekly Pattern:** Saturday/Friday are peak days (2x weekday volume)
- Schedule major promotions on **Fri-Sat** to maximize impact
- Increase promotional frequency — product shows extreme price sensitivity
- Test deeper discounts to drive even higher volumes

4. Year-over-Year Decline

- Sales dropped **-15.1%** from 2014→2015, **-26.7%** from 2015→2016
- Volume declining: from 6,176 units/day (2014) to 4,033 units/day (2016)
- Investigate market factors, competition, or product lifecycle stage